Ashutosh Tiwari

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Work Experience

Senior Machine Learning Engineer (ML Platform & Frameworks) Jul. 2024 – Present
ADOBE FIREFLY *GENERATIVE AI, COMPUTER VISION, LARGE LANGUAGE MODELS*

- Leading part of ML Platform focused on offline inference for videos/images and on training frameworks for foundation models.
- **Distributed Inference Frameworks**: Designed and led engineering for the org's online and offline inference systems powering experimentation and production at scale.
  - Built online inference on vLLM + Ray for rapid experimentation with LLMs such as Falcon-40B and Llama 2 70B.
  - Built offline inference on Ray Data + PyTorch Lightning, achieving 10–50× faster image loading and 3–8K images/sec on 32 GPUs.
- **Foundation Model Training Framework**: Leading development of a PyTorch-based training framework adopted across the org for training generative models on billions of images and videos.
  - Designed a plugin-based strategy pattern over FSDP/FSDP2 to make distributed training strategies pluggable.
  - Integrated Flash Attention 2/3, context parallelism, and distributed attention for DiT text-to-image and text-to-video training.
- **Enrichment Service**: Orchestration and abstraction layer over the inference framework and other ML Platform offerings; runs fire-and-forget pipelines on datasets of billions of images and videos.
- **Data Quality Framework**: Wrote the org's first data quality framework on Cerberus, used across enrichment pipelines to validate the correctness of feature-generation outputs.

Senior Software Engineer (Data Platform) Sep. 2023 – Jul. 2024
EVOLUTIONIQ *DATA PLATFORM, NATURAL LANGUAGE PROCESSING, REGRESSION*

- Part of the Scoring and Quality sub-team, working at the intersection of generative AI, fin-tech, and the health sector.
- **Data Pipelines**: Led development of pipelines that ingest and process the data used for training across all of our machine learning models.
- **Model Training Pipelines**: Led pipelines that train models to predict a claimant's expected time to return to work, ICD extraction from diagnosis notes, and alternate VOC recommendations.
- **Fairness & Bias Framework**: Built a framework to evaluate and measure model fairness and bias metrics across different demographic groups.

Software Dev Engineer II (ML Platform) Jan. 2019 – Jul. 2021
SWIGGY *DATA SCIENCE PLATFORM, TIME SERIES FORECASTING*

- **Online Inference Platform**: Core contributor to the online inference platform serving 18M+ monthly transacting users across food delivery and quick commerce, with models autoscaled by real-time load metrics.
  - Engineered a dynamic request-routing layer that evaluated incoming API calls against geo-tags, traffic profiles, and service metadata, then dispatched them to model clusters partitioned by region and load domain.
  - Implemented routing/dispatch on an Akka-style actor framework for resilient asynchronous invocation, fault isolation across clusters, and backpressure-aware scheduling at sustained low P99 under high QPS.
- **Feature Store**: Built and maintained a Feature Store and pipeline serving on-demand features to deployed ML models at production scale (4Bn rows, 10K QPS), with multichannel ingestion via Spark, Flink, and user files.
- **Forecasting & Correlation Platform**: Founding member of the platform used by many teams to forecast time series; forecasts powered critical scaling decisions across the org in real time.
- **DAQ**: Led DAQ, a tool used to scrape APIs at scale, collecting 15M rows daily for analysis and model training.

Software Development Engineer (Search Relevance) Sep. 2017 – Jan. 2019
FLIPKART (a Walmart company) *SEARCH RELEVANCE, QUERY INTENT, NLP*

- **Search Intent Models**: Built and improved CRF/Neural Network-based intent models powering user search and discovery for millions every day; identified error classes, designed solutions, and shipped fixes.
- **Tail Query Classifier**: Implemented a FastText-based query store classifier that predicts the category of a tail query.
- **ML Workflow Automation**: Built the first workflow at Flipkart to automate training and auto-deployment of search models—written first in Luigi and later migrated to Airflow.
  - Wrote a generic Airflow framework that creates DAGs at runtime for different ML models and orchestrates the train→deploy flow, including data and model validations.
- **Large-Scale Data Pipelines**: Implemented Cascading/HDFS pipelines (4Bn+ datapoints) to extract user-event data and transform it for model training.

Software Development Engineer Sep. 2016 – Sep. 2017
GROUPON *BACKEND ENGINEERING*

- **Cyclops**: Worked on Cyclops, an interface between Customer Representatives and internal services; deployed to all countries in which Groupon operates.

Software Engineer Sep. 2015 – Aug. 2016
NETSPEED SYSTEMS (Acquired by Intel) *GRAPH ALGORITHMS, NETWORK ON CHIP*

- **Network-on-Chip Modules**: Led engineering on modules including Polarity-based Arbitration, Multi-Cast Filtering, and Structural Latency Breakdown.

Publications

Accepted at *NetSci 2023* (Poster Presentation) and *IC2S2 2023* (Parallel Talk) as **first author**

[1] **Ashutosh Tiwari**, Prof. Sadamori Kojaku, Prof. Yong-Yeol Ahn, "Biased Contrastive Learning debiases Graph Neural Networks," *International Conference on Network Science (NetSci)*, 2023. Poster

In this work, we propose a non-parametric contrastive learning framework to learn debiased graph embeddings with respect to sensitive node attributes and structural homophily. Through empirical evaluations on different datasets, we demonstrate that our method offers a better approach to debiasing compared to existing approaches and thus results in more organic recommendations across different GNN architectures.

Education

Indiana University, Bloomington — Bloomington, IN
Master of Science in Computational Data Science 3.87/4.0 — *Aug. 2021 – May 2023*

National Institute of Technology — Patna, India
Bachelor of Science in Computer Science 8.32/10.0 — *Aug. 2011 – May 2015*

Research Experience / Independent Study

- Implementing novel training frameworks that result in unbiased (or fair) recommendation systems from "biased" datasets at IUNI with Prof. YY Ahn and Prof. S Kojaku as part of my Independent Study. Summer 2022 - Summer 2023
- Working as a paid RA for, **"User Intent as a Network"** with Prof. YY Ahn, P Kantak and FB Yara. Project is funded by Kelly Business School. Goal is to be able to quantify and thus act upon that "intent" network. Summer 2022 - Fall 2022
- Was part of NLP Lab@IUB with Prof. D Cavar. Contributed to design of TieML and Events' Timeline modelling using different fine-tuned Large Language Models (LLMs). Fall 2021

Teaching Experience

- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2023
- Teaching Assistant for Machine Learning (CSCI-B 555) with Prof. R Khardon — Fall 2022
- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2022

Selected Projects

BiasNet | *Deep Reinforcement Learning, PyTorch, Actor Critic Algorithm, On-policy Model Free* — Code/Report
- Learning to fight in Street Fighter II with induced relational bias from differential game scenes.

DeepFoodie | *Python, Tensorflow, Self Supervised Deep Clustering, Deep Learning, Transfer Learning* — Code/Report
- Clustering dishes on basis of their ingredient embeddings. These ingredient embeddings are generated by a NN.

BlindNet | *Python, Pytorch, Deep Learning, Transfer Learning* — Code/Report
- Image to vector generation on Coco dataset.

Continuous Dominant Set Repair | *C++, Graph Algorithms, Guha and Khuller's Algo., Greedy* — Code
- Repairs a broken link in Continuous Dominant Set in $O(\Delta^2)$, where Δ being the avg cardinality of connected graph.

Humana Mays Healthcare Analytics Case Competition | *Boosted Trees, Feature Engineering* — Code
- 11th rank on leaderboard. Hosted by TAMU and Humana Mays, 2021.

AnalyticsVidhya Solutions | *Python, Tensorflow, Torch, Time Series, Feature Engineering, Catboost* — Code

Investigating Bias Manifolds | *Bias Manifolds, Bias Progression, Measuring Bias, Python, Word2vec* — Code/Report

Technical Skills

Search Relevance, Recommendations, Graph Neural Networks, Fairness Aware Modeling, Computational ML, NLP, Computer Vision, DL, Spark, Python, Scala, C++, Contrastive Learning, Large Language Models
Frameworks/Libraries/Tools: Ray, Pytorch Geometric, Pytorch Lightning, Pytorch, Tensorflow, Sklearn, Numpy, Pandas, Matplotlib, HDFS, Spark, Kafka, Flink, AWS Sagemaker, DynamoDB, Faiss, vLLM, Django